Maximum Flow Analysis

Network Capacity, Optimal Routing, and Cut Theory

1. The Maximum Flow Problem

Maximum Flow involves finding the greatest possible amount of "traffic" (data, water, electricity) that can pass from a **Source (s)** to a **Sink (t)** in a directed graph, subject to capacity constraints on each edge.

> **Conservation of Flow:** For every node except the source and sink, the amount of flow entering must exactly equal the amount of flow leaving.

2. Essential Terminology

- **Capacity:** The maximum allowable flow on an edge (e.g., a 10Gbps cable).
- **Residual Capacity:** The original capacity minus the current flow. It represents how much more can be sent.
- **Augmenting Path:** A path from source to sink in the residual network where every edge has residual capacity > 0.
- **Super-Source/Sink:** Artificial nodes used to simplify problems with multiple starting or ending points.

3. Residual Networks & Reversed Edges

Maximum flow algorithms use a **Residual Network** to keep track of available capacities. A critical component is the **Reversed Edge**.

When flow is sent forward, we add capacity to a virtual reversed edge. This allows the algorithm to "undo" or reroute flow if it finds a more efficient path later, effectively pushing flow back to optimize the global network.

4. The Max-Flow Min-Cut Theorem

This is one of the most powerful theorems in optimization. It states that the **Maximum Flow** possible in a network is exactly equal to the capacity of the **Minimum Cut**.

> **A Cut:** A partition of vertices into two sets (S and T) where the source is in S and the sink is in T. The capacity of the cut is the sum of capacities of edges crossing from S to T.

Identify the minimum cut to find the **bottleneck** of the system. This tells engineers exactly where a system must be upgraded to increase total throughput.

5. Mathematical Constraints

The problem is defined by three fundamental rules:

1. **Capacity Constraint:** $0 \le f(u,v) \le c(u,v)$.
2. **Skew Symmetry:** $f(u,v) = -f(v,u)$.
3. **Flow Conservation:** $\Sigma f(u,w) = 0$ (for all nodes $u \neq s,t$).

Industrial Applications (Part 1)

Finding the maximum flow isn't just a theoretical exercise; it is the core of modern logistics. For example, airline scheduling uses flow networks to determine how to route aircraft between hubs to maximize passenger throughput while respecting seating capacities (the edge capacity).

> **Bottleneck Analysis:** By finding the Minimum Cut, a factory manager can identify exactly which conveyor belt or machine is limiting the entire plant's production rate.

The Super-Source Technique

In real-world networks with multiple factories (sources) and multiple distribution centers (sinks), we transform the complex graph into a single-source single-sink problem. By adding one "Super-Source" node connected to all factories with infinite capacity, we allow standard Ford-Fulkerson logic to solve multi-point logistical challenges.

Industrial Applications (Part 2)

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